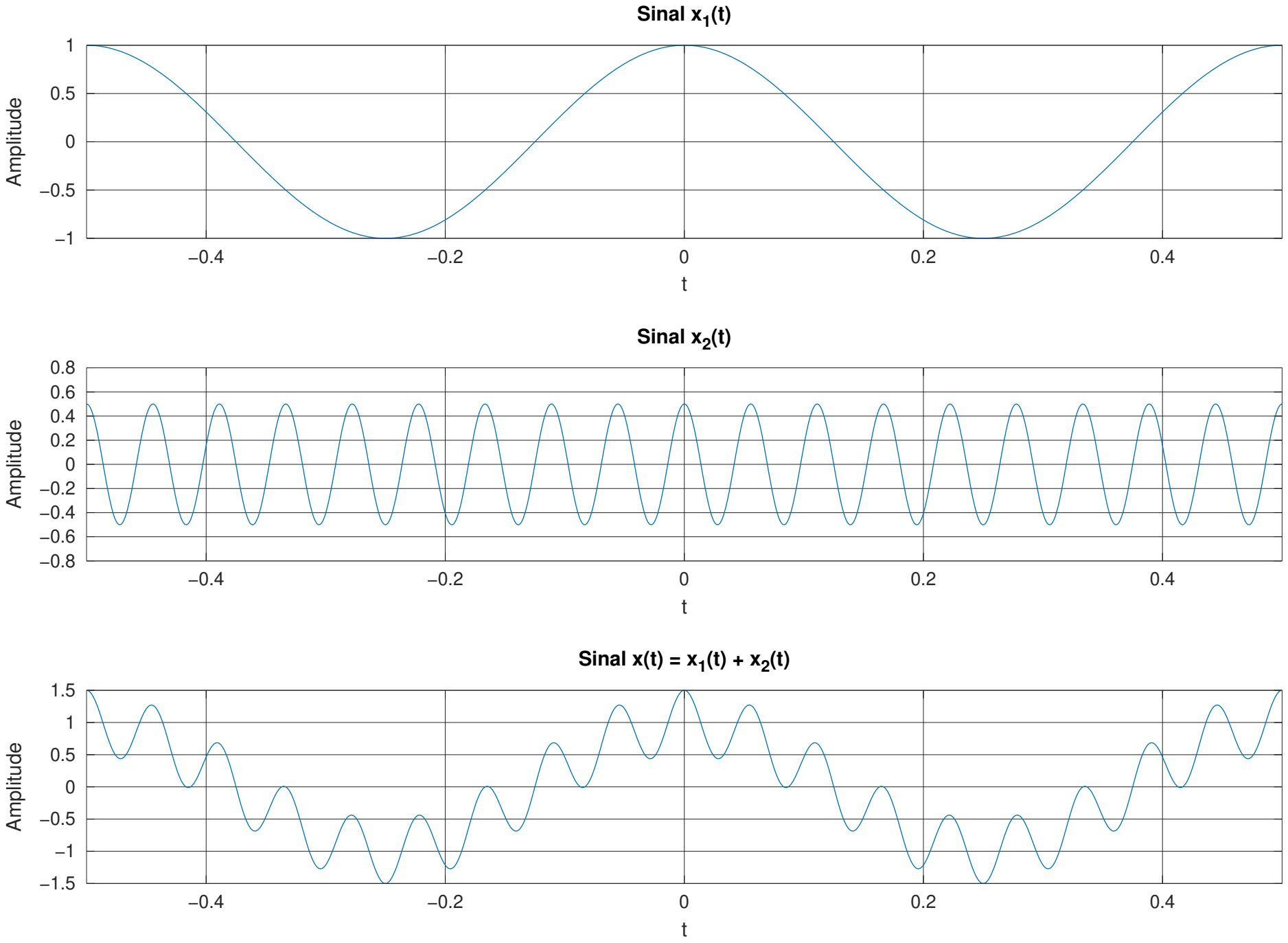
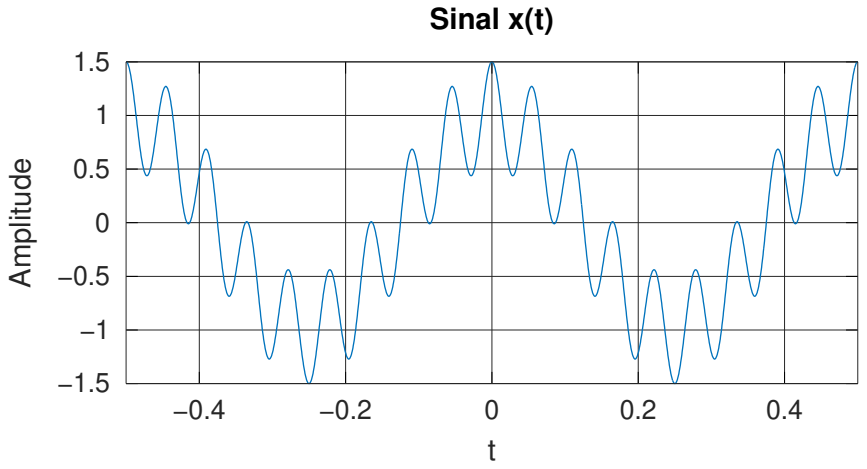
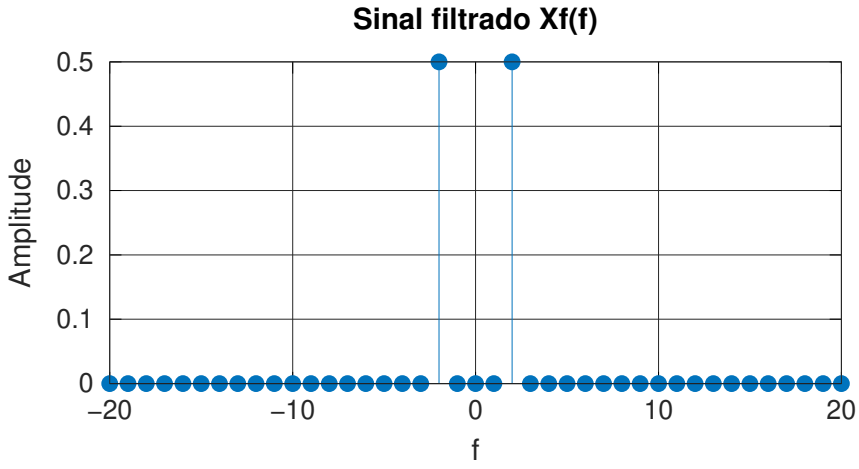
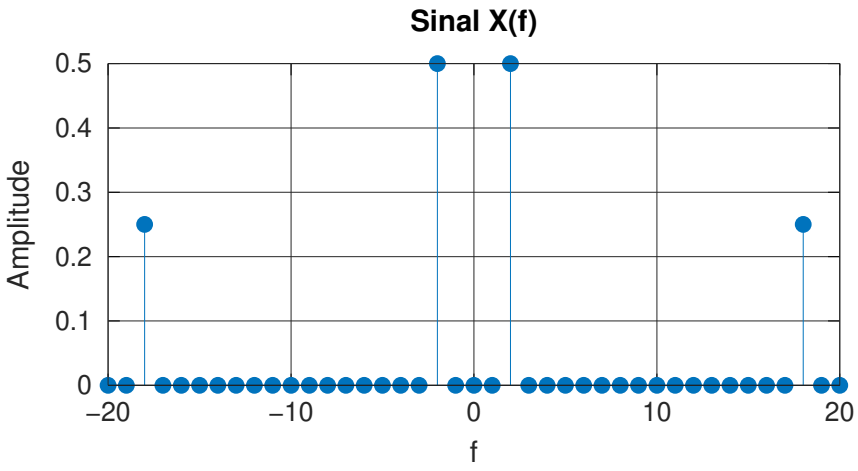
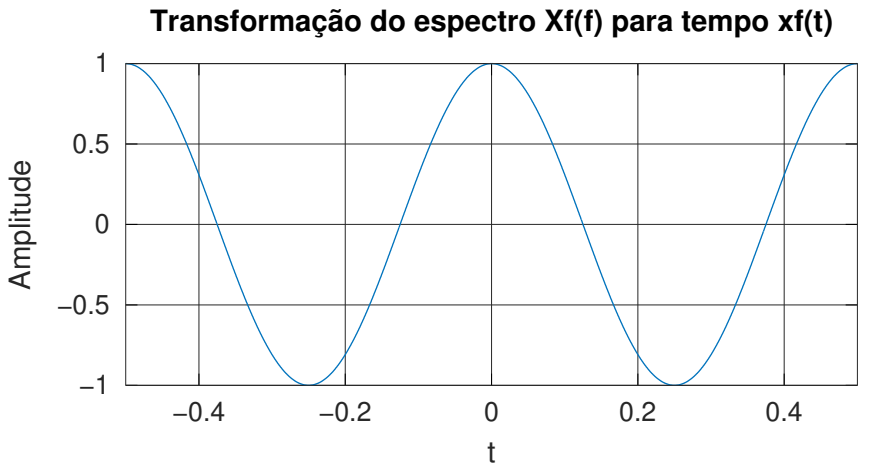
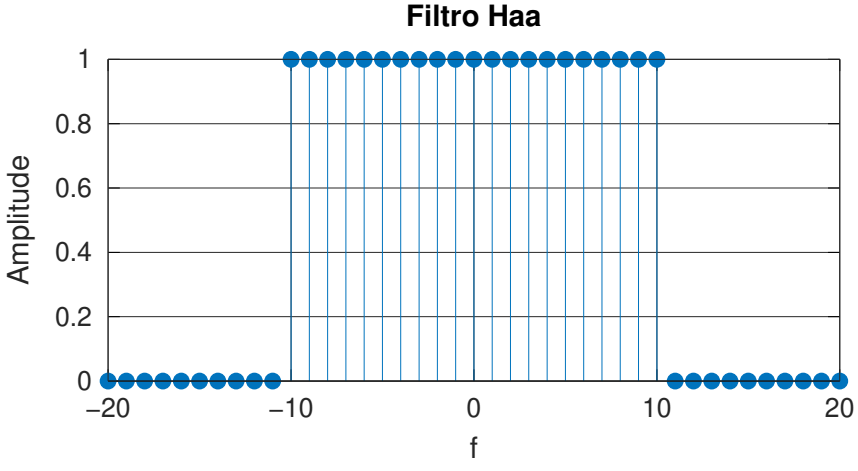

Sinal $x_1(t)$
Amplitude
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4
t
Sinal $x_2(t)$
Amplitude
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−0.6
−0.8
−0.4
−0.2
0
0.2
0.4
t
Sinal $x(t) = x_1(t) + x_2(t)$
Amplitude
1.5
1
0.5
0
−0.5
−1
−1.5
−0.4
−0.2
0
0.2
0.4
t

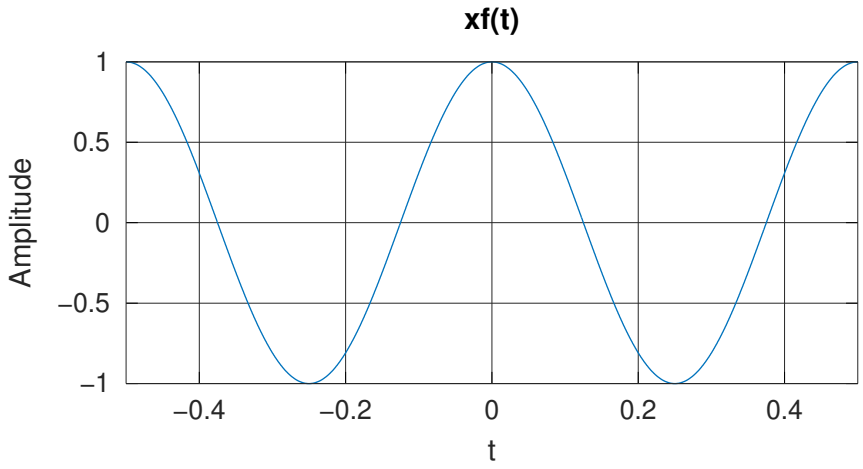

Sinal x(t)
Amplitude
t
1.5
1
0.5
0
−0.5
−1
−1.5
−0.4
−0.2
0
0.2
0.4

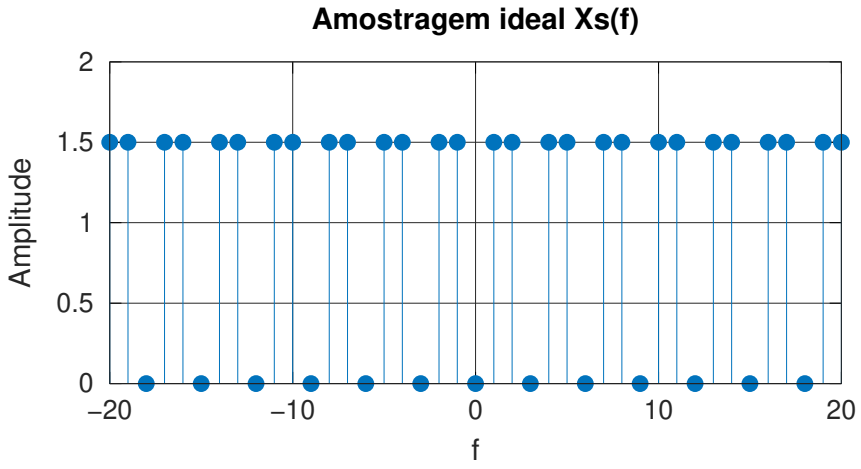

Sinal filtrado Xf(f)
Amplitude
f
0.5
0.4
0.3
0.2
0.1
0
−20
−10
0
10
20

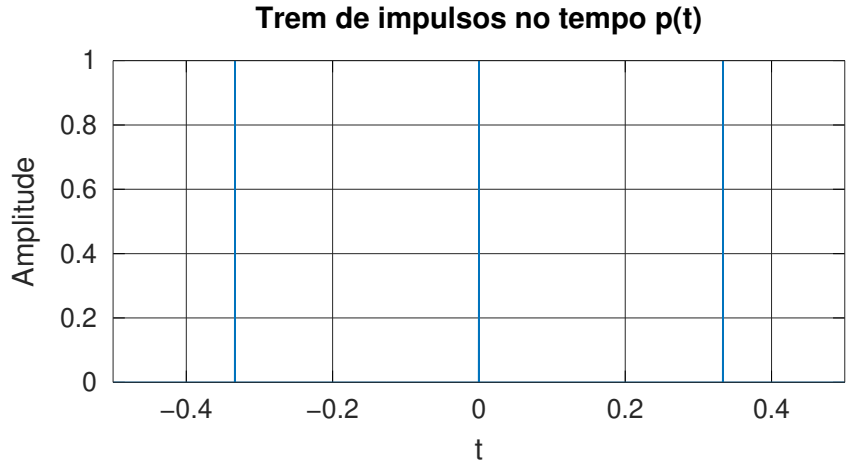

Sinal X(f)
Amplitude
f
0.5
0.4
0.3
0.2
0.1
0
−20
−10
0
10
20

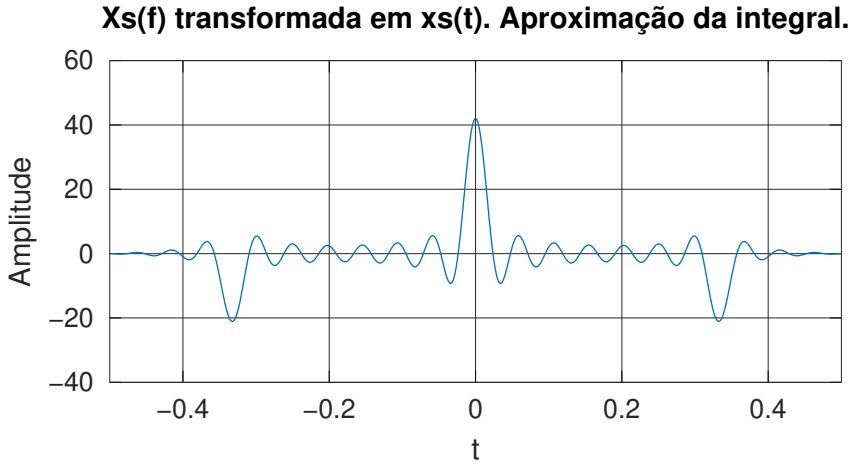

Transformação do espectro Xf(f) para tempo xf(t)
Amplitude
t
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

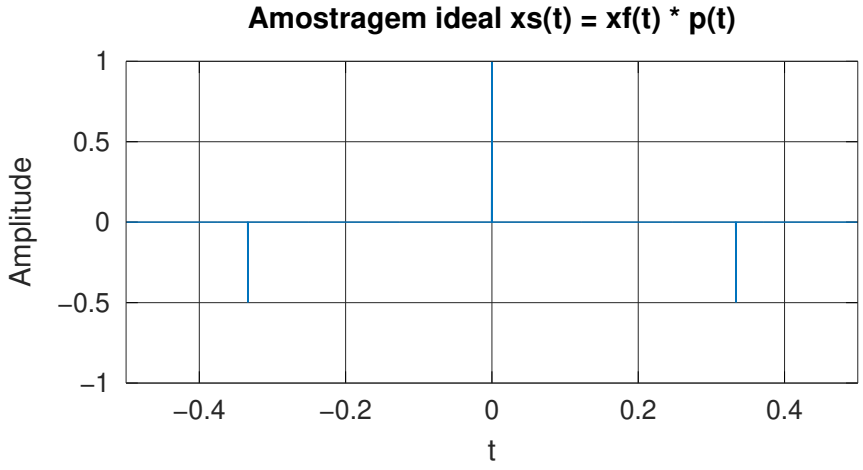

Filtro Haa
Amplitude
f
1
0.8
0.6
0.4
0.2
0
−20
−10
0
10
20

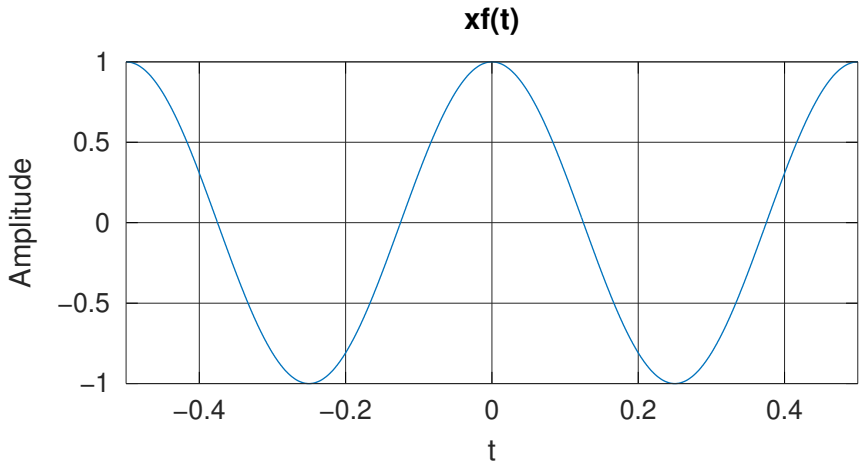

xf(t)
Amplitude
t
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

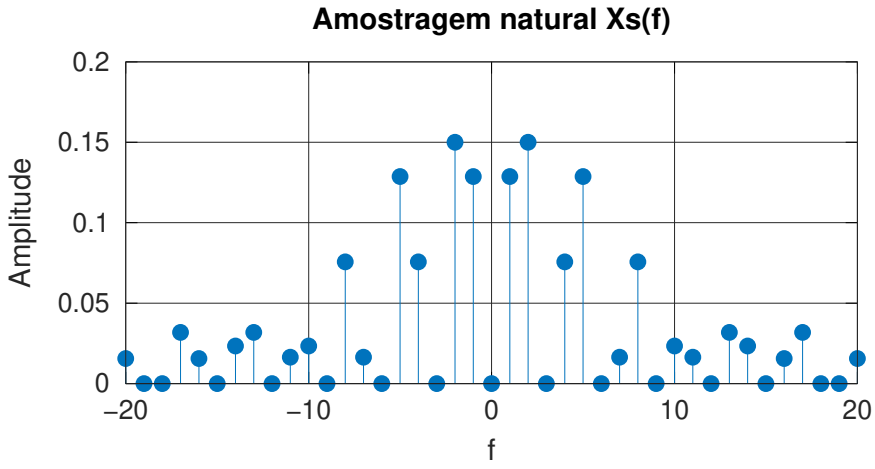

Amostragem ideal Xs(f)
Amplitude
f
2
1.5
1
0.5
0
−20
−10
0
10
20

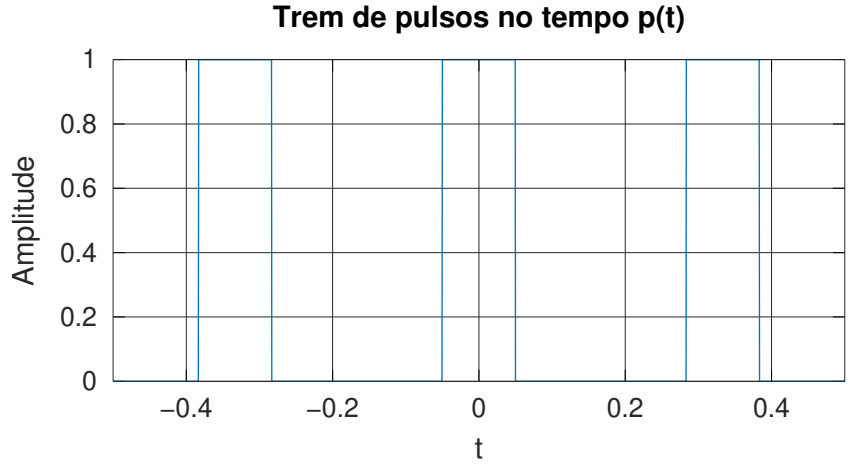

Trem de impulsos no tempo p(t)
Amplitude
t
1
0.8
0.6
0.4
0.2
0
−0.4
−0.2
0
0.2
0.4

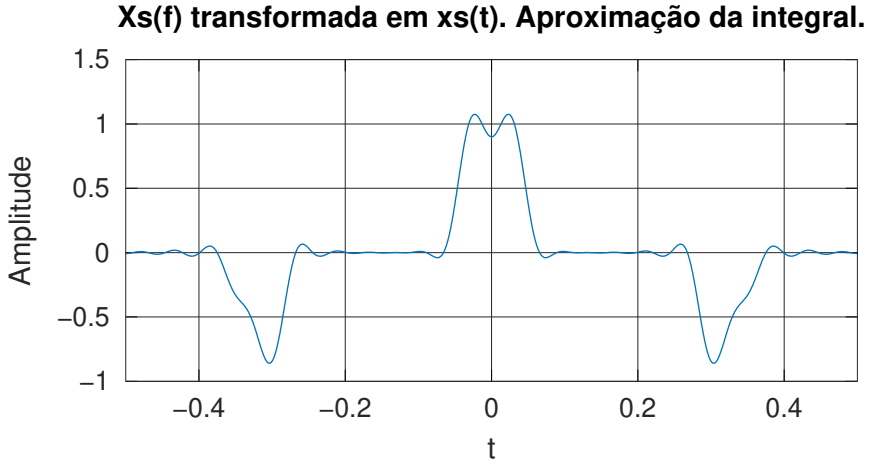

Xs(f) transformada em xs(t). Aproximação da integral.
Amplitude
t
60
40
20
0
−20
−40
−0.4
−0.2
0
0.2
0.4

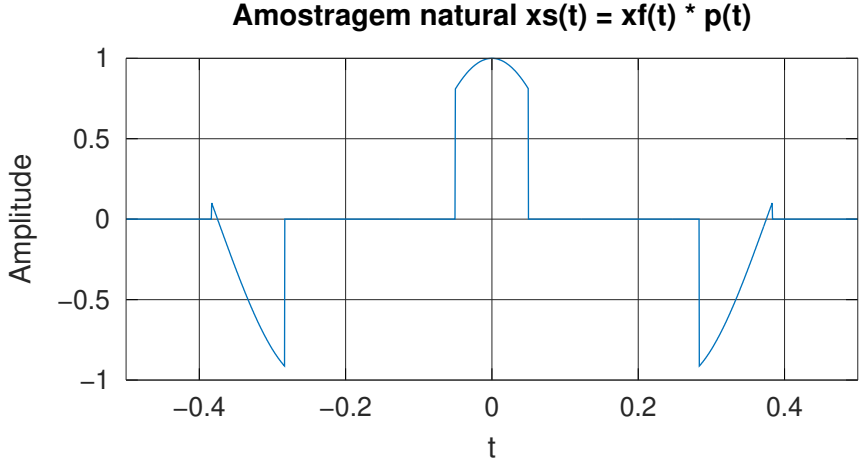

Amostragem ideal xs(t) = xf(t) * p(t)
Amplitude
t
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

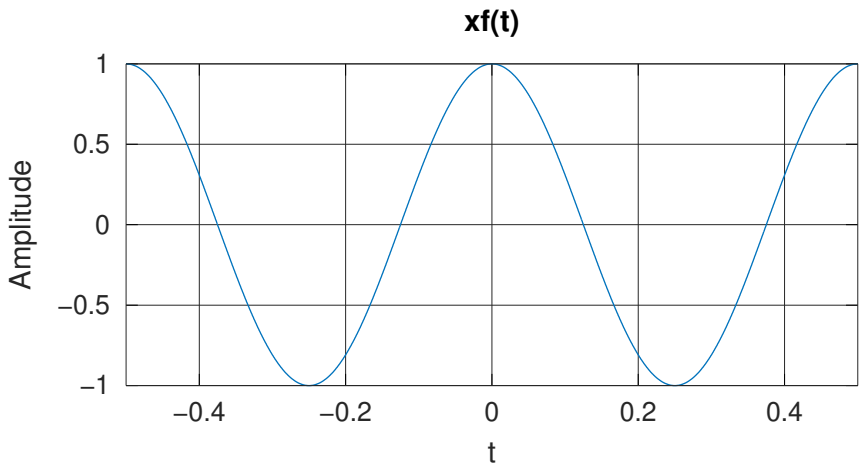

xf(t)
Amplitude
t
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

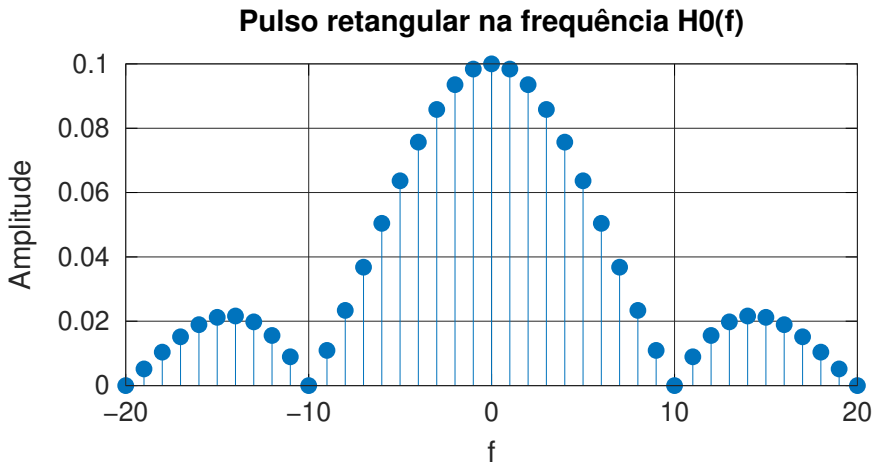

Amostragem natural Xs(f)
Amplitude
f
0.2
0.15
0.1
0.05
0
−20
−10
0
10
20

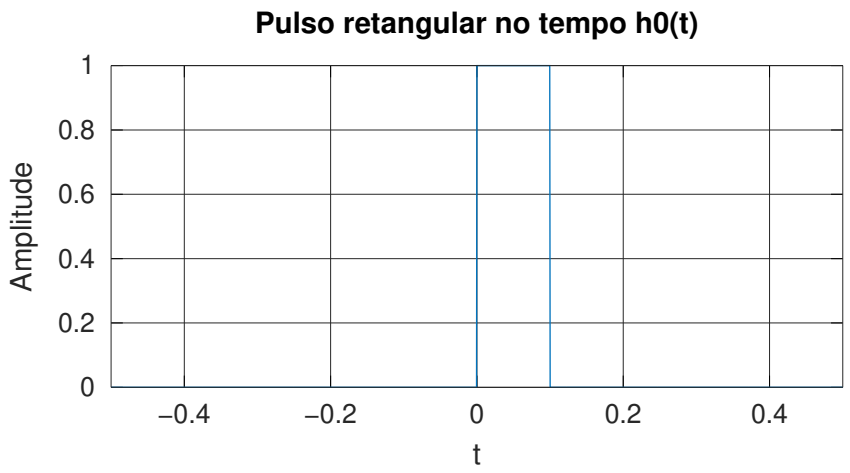

Trem de pulsos no tempo p(t)
Amplitude
t
1
0.8
0.6
0.4
0.2
0
−0.4
−0.2
0
0.2
0.4

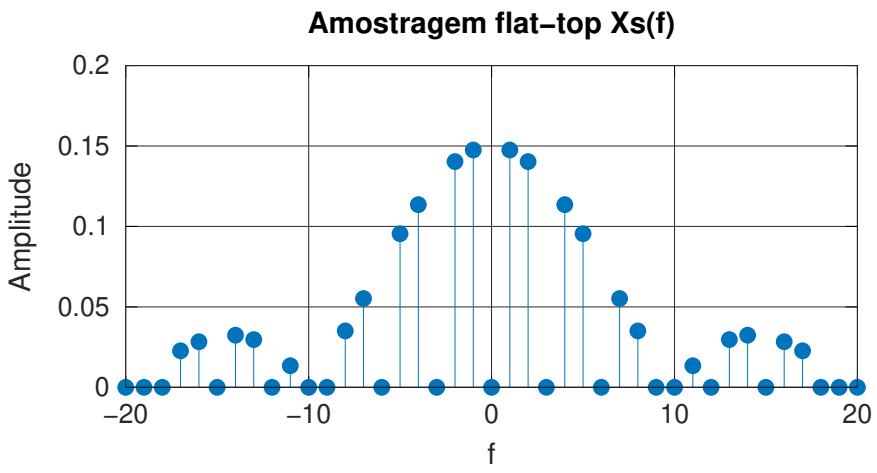

Xs(f) transformada em xs(t). Aproximação da integral.
Amplitude
t
1.5
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

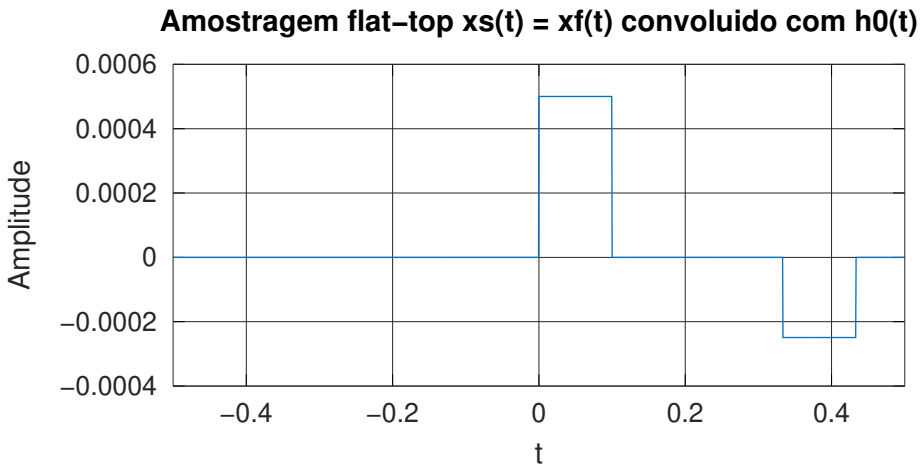

Amostragem natural xs(t) = xf(t) * p(t)
Amplitude
t
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

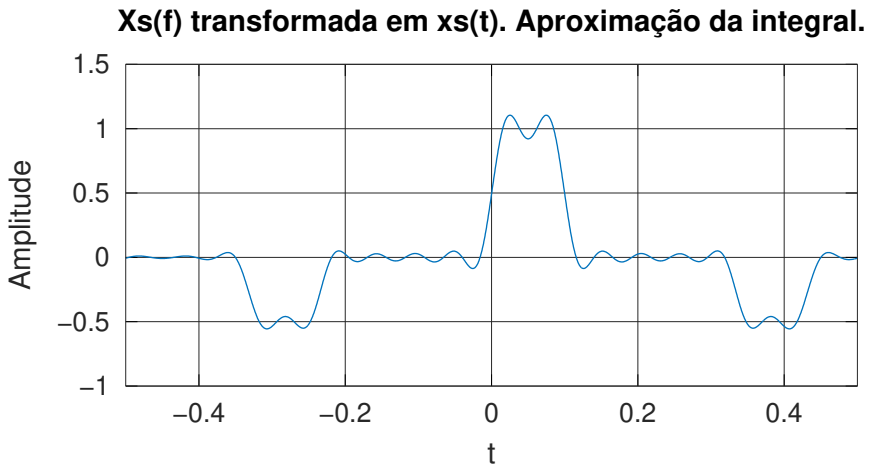

xf(t)
Amplitude
t
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

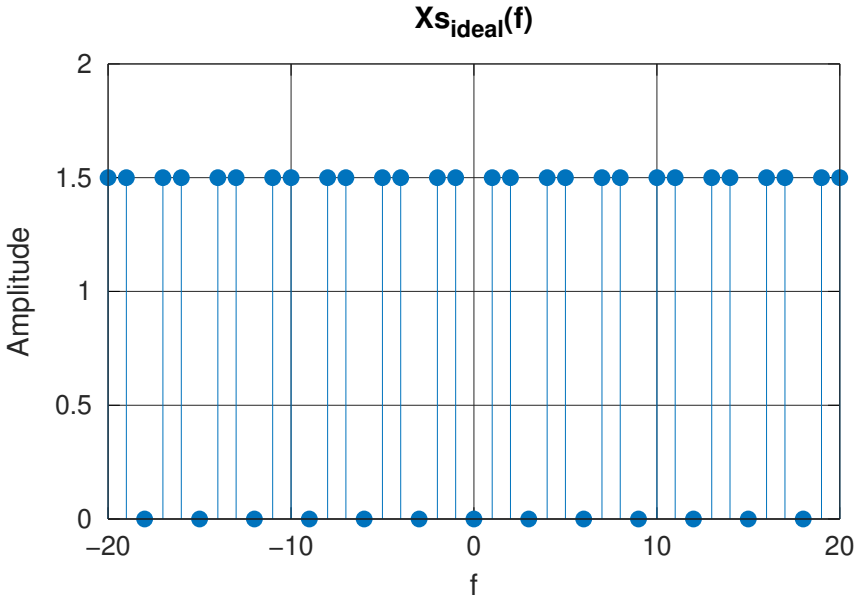

Pulso retangular na frequência H0(f)
Amplitude
f
0.1
0.08
0.06
0.04
0.02
0
−20
−10
0
10
20

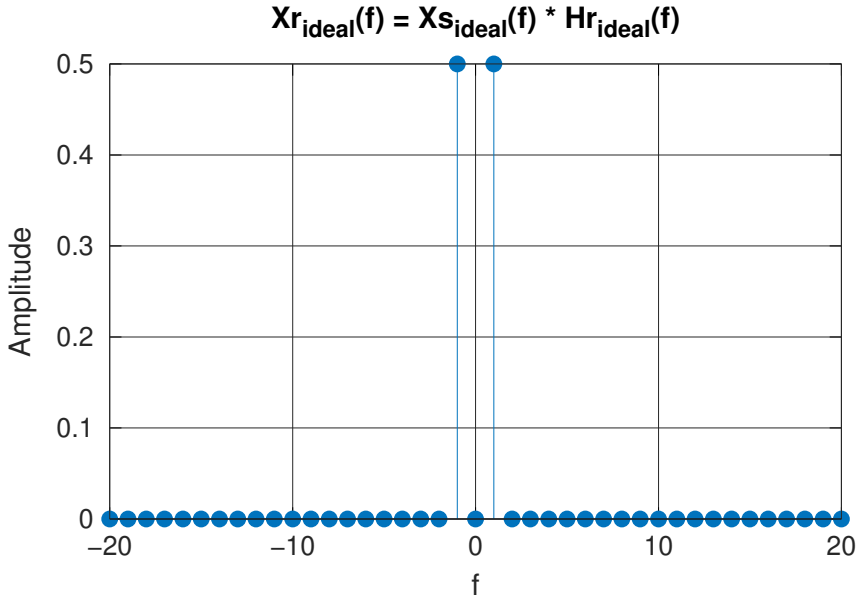

Pulso retangular no tempo h0(t)
Amplitude
t
1
0.8
0.6
0.4
0.2
0
−0.4
−0.2
0
0.2
0.4

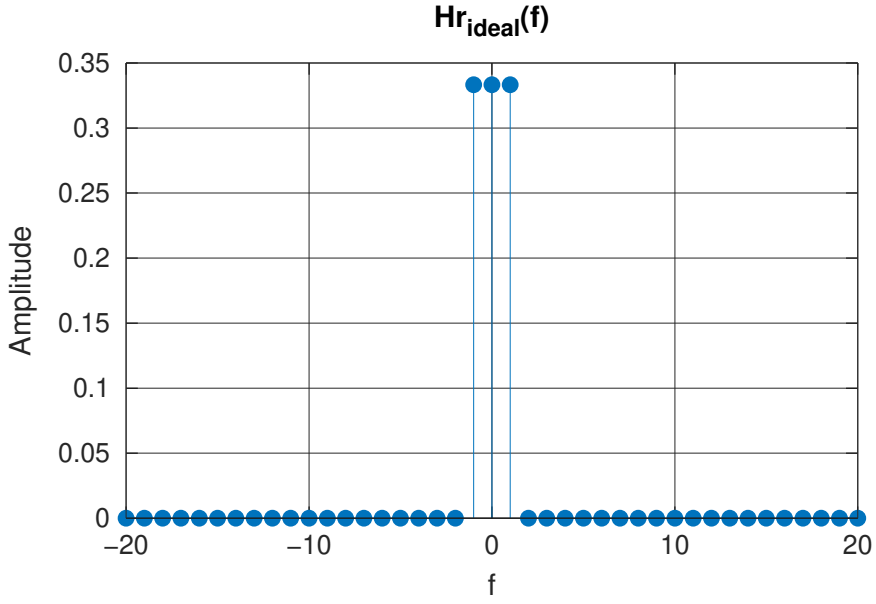

Amostragem flat−top Xs(f)
Amplitude
f
0.2
0.15
0.1
0.05
0
−20
−10
0
10
20

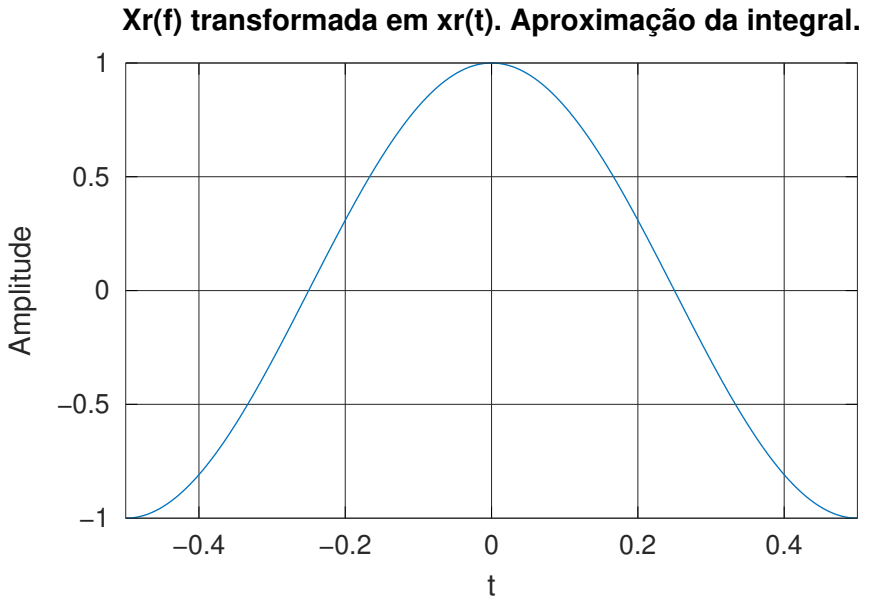

Amostragem flat−top xs(t) = xf(t) convoluido com h0(t)
Amplitude
t
0.0006
0.0004
0.0002
0
−0.0002
−0.0004
−0.4
−0.2
0
0.2
0.4

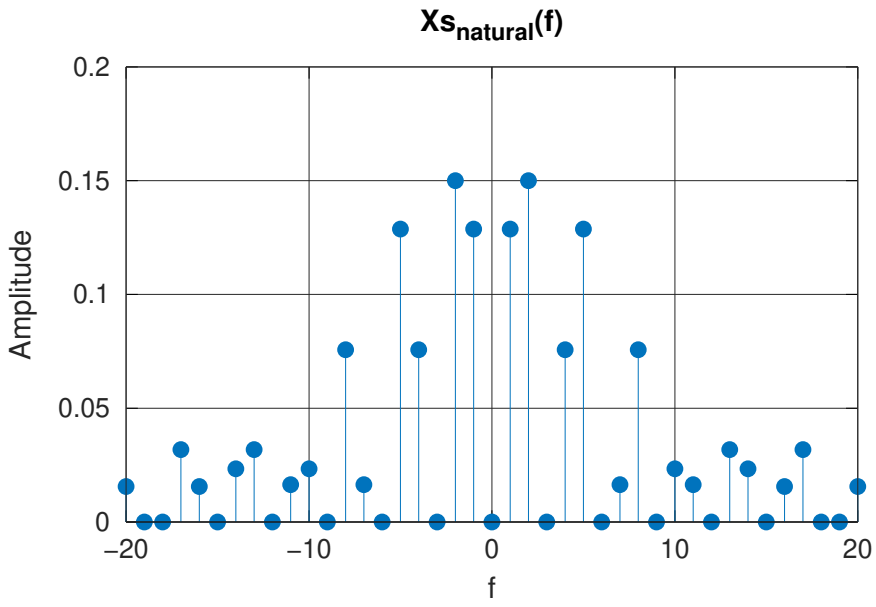

Xs(f) transformada em xs(t). Aproximação da integral.
Amplitude
t
1.5
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

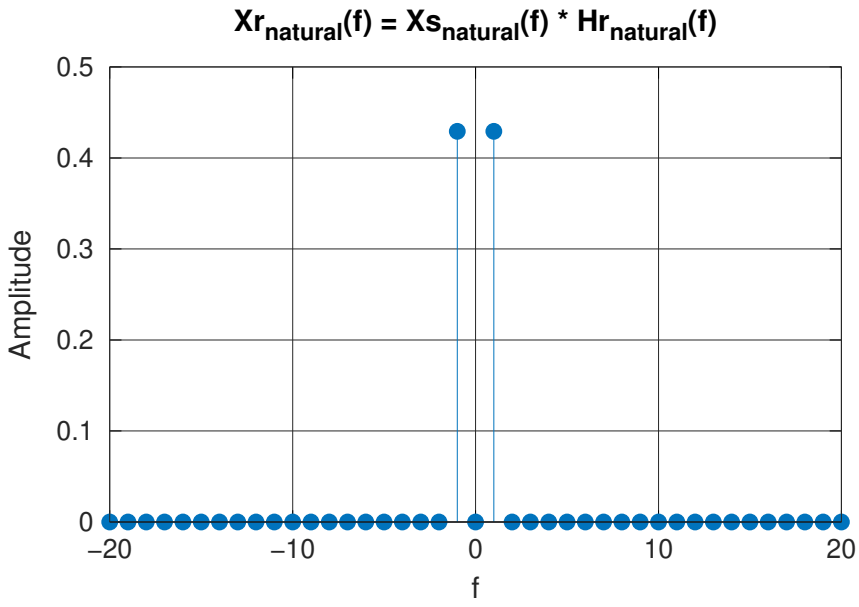

Xs_ideal(f)
Amplitude
f
2
1.5
1
0.5
0
−20
−10
0
10
20

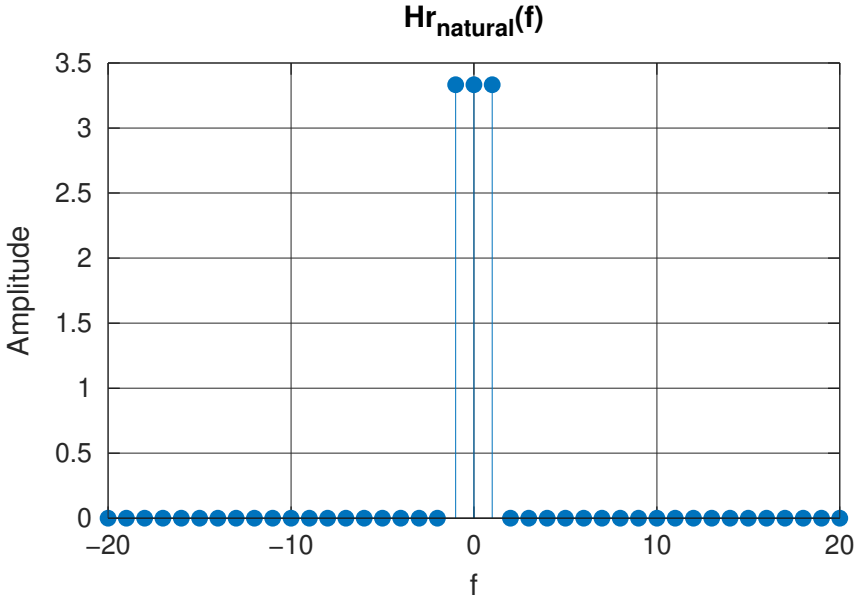

Xr_ideal(f) = Xs_ideal(f) * Hr_ideal(f)
Amplitude
f
0.5
0.4
0.3
0.2
0.1
0
−20
−10
0
10
20

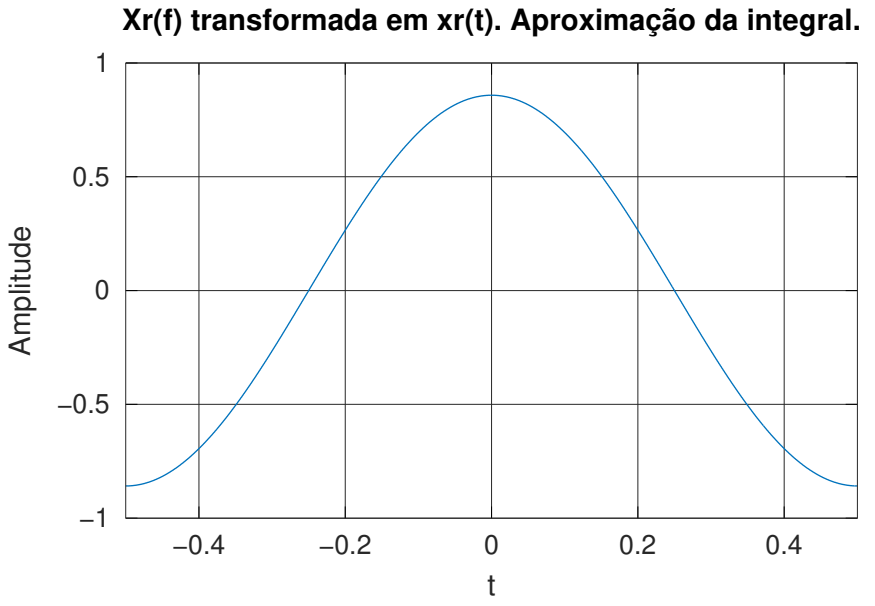

Hr_ideal(f)
Amplitude
f
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
−20
−10
0
10
20

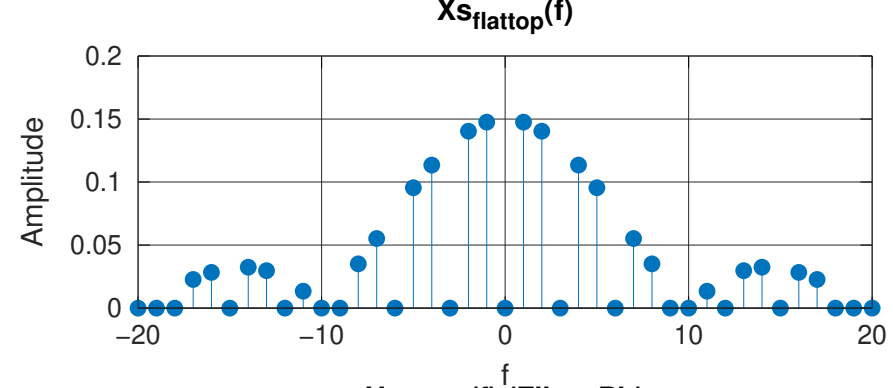

Xr(f) transformada em xr(t). Aproximação da integral.
Amplitude
t
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

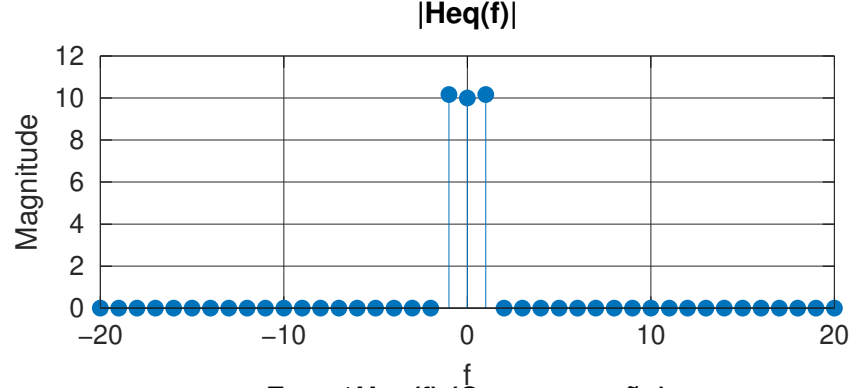

Xs natural(f)
Amplitude
f
0.2
0.15
0.1
0.05
0
−20
−10
0
10
20

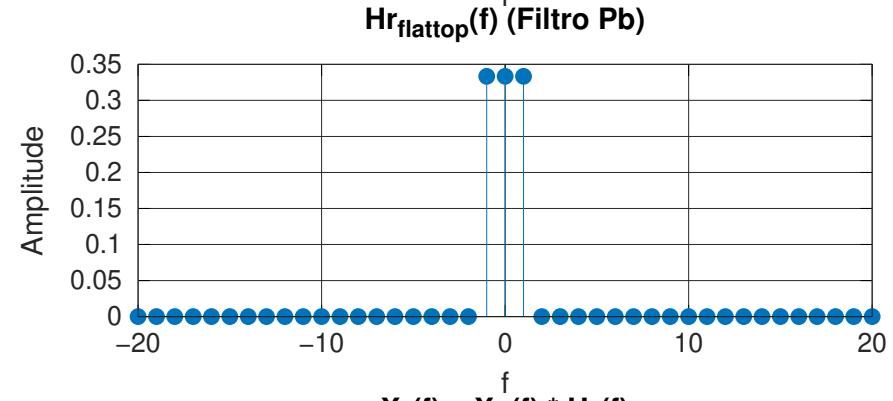

Xr natural(f) = Xs natural(f) * Hr natural(f)
Amplitude
f
0.5
0.4
0.3
0.2
0.1
0
−20
−10
0
10
20

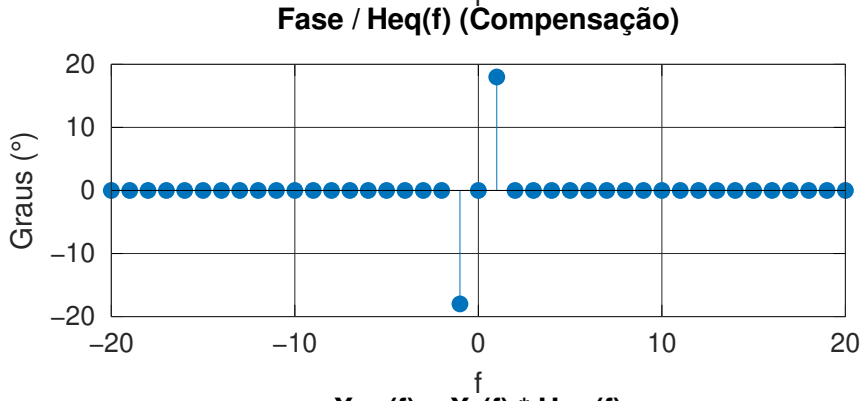

Hr natural(f)
Amplitude
f
3.5
3
2.5
2
1.5
1
0.5
0
−20
−10
0
10
20

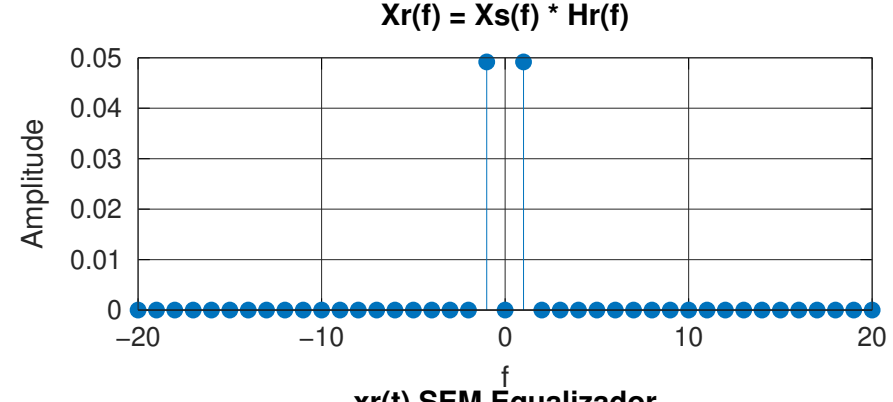

Xr(f) transformada em xr(t). Aproximação da integral.
Amplitude
t
1
0.5
0
−0.5
−1
−0.4
−0.2
0
0.2
0.4

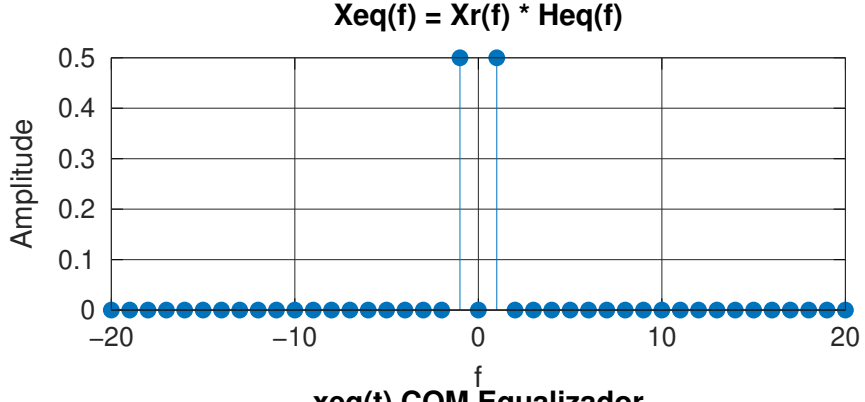

Xs_flattop(f)
Amplitude
f

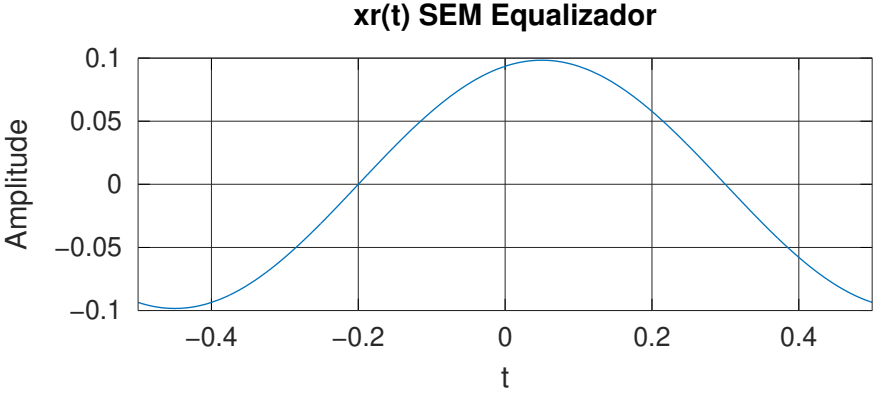

|Heq(f)|
Magnitude
f

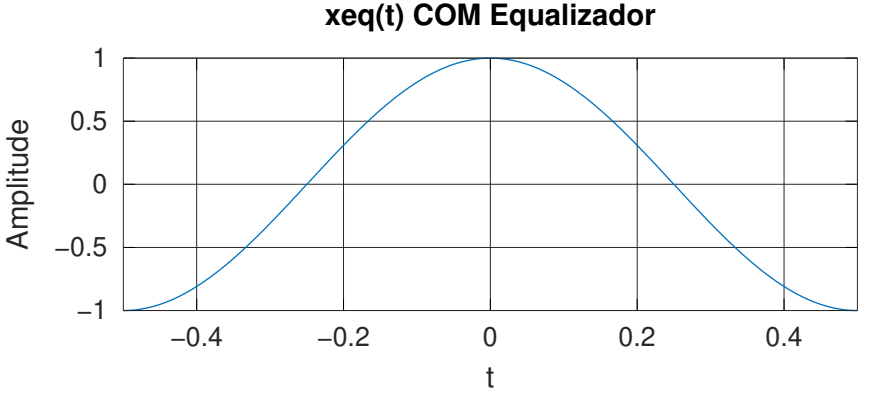

Hr_flattop(f) (Filtro Pb)
Amplitude
f


Fase / Heq(f) (Compensação)
Graus (°)
f


Xr(f) = Xs(f) * Hr(f)
Amplitude
f


Xeq(f) = Xr(f) * Heq(f)
Amplitude
f


xr(t) SEM Equalizador
Amplitude
t


xeq(t) COM Equalizador
Amplitude
t